IIT ROORKEE

FREE ONLINE EDUCATION swayam शिक्षित भारत, उन्नत भारत

NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-53

DC Charging (CCS2)-IV

Dr. Apurv Kumar Yadav

Department of Electrical Engineering



Recap

- CCS2 charging sequence is being studied.
- SLAC Procedure to logically connect the PLC modems of EV and EVSE which are making physical connection.

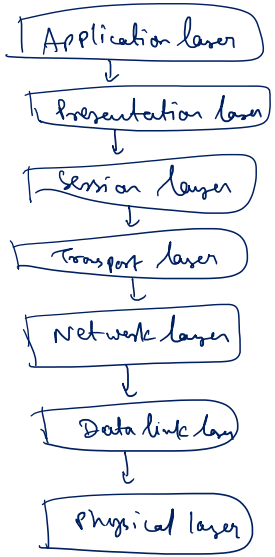
CCS2 HLC

- The CCS2 high level communication (HLC) happens over PLC (on the CP-PE lines).
- HLC is used for sending/receiving charging parameters and information
- The messages are exchanged following the 7 OSI layer model, which is used to perform data transfer between two computers connected via network.
- For each message the identified OSI Layers , beginning with the physical connection and proceeding step by step to the control application.

Source: 1. https://morth.gov.in/sites/default/files/ASI/320201825106PMAIS_138_Part_2_F.pdf
2. https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

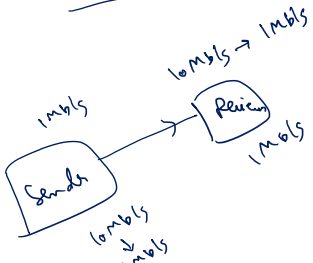
7 OSI (Open system interconnection) Layer

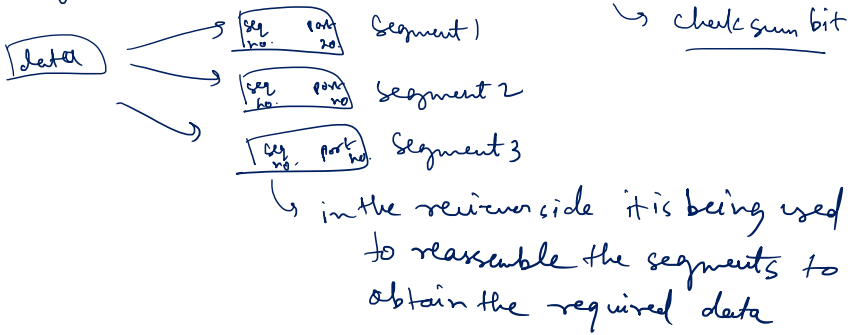
- Each layer consists of package of protocols which are needed to perform specific operation to complete successful communication between computers having different architectures.



7 OSI Layer

1. Application layer → web surfing → HTTP / HTTPS ; file → FTP ;
   ↳ Consists of protocols which is being used by the network application, to perform required user activity
2. Presentation layer → Translation, data compression, Encryption/Decryption
3. Session layer → authetication, authorization, session management
4. Transport layer → Segmentation, Flow control, Error control
   ↳ checksum bit



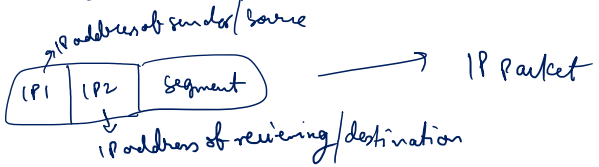


↳ in the reciever side it is being used to reassemble the segments to obtain the required data

7 OSI Layer

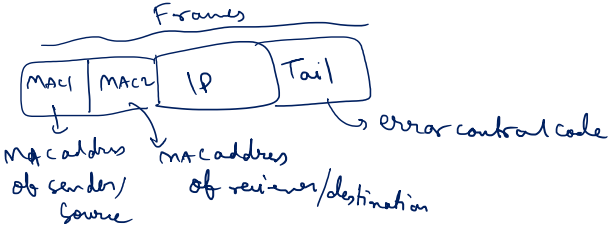
Data → Segments → IP Packets → Frames

(5) Network layer → logical addressing, Routing, Path determination
- → IP addressing
- → Routing



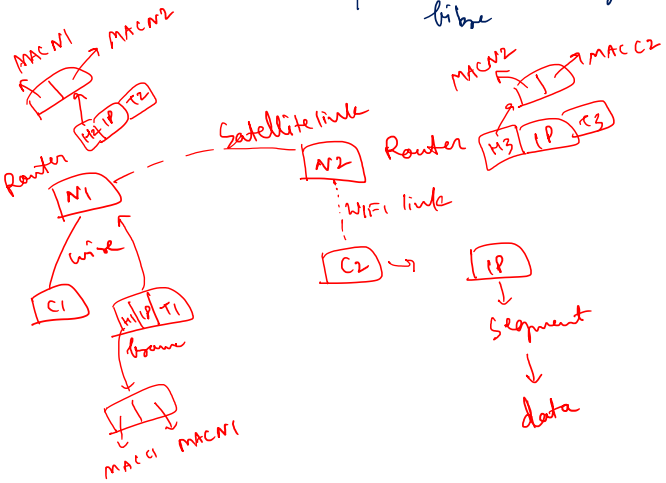
(6) Data link layer → Physical addressing, control how the data is being placed/taken from the physical medium
- → MAC (media access control) address
  - it is a 12 digit alphanumeric number

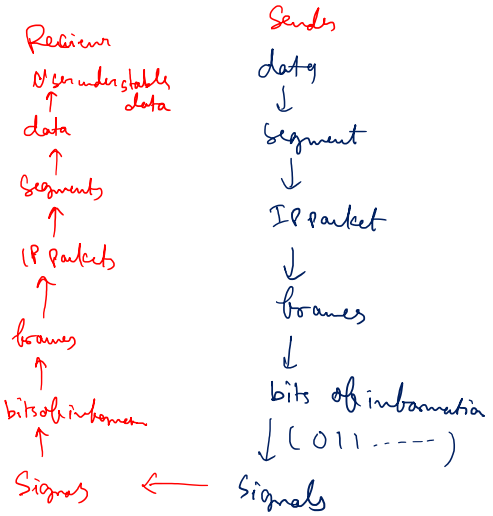
physical media
- → optical fiber
- → copper wire
- → air



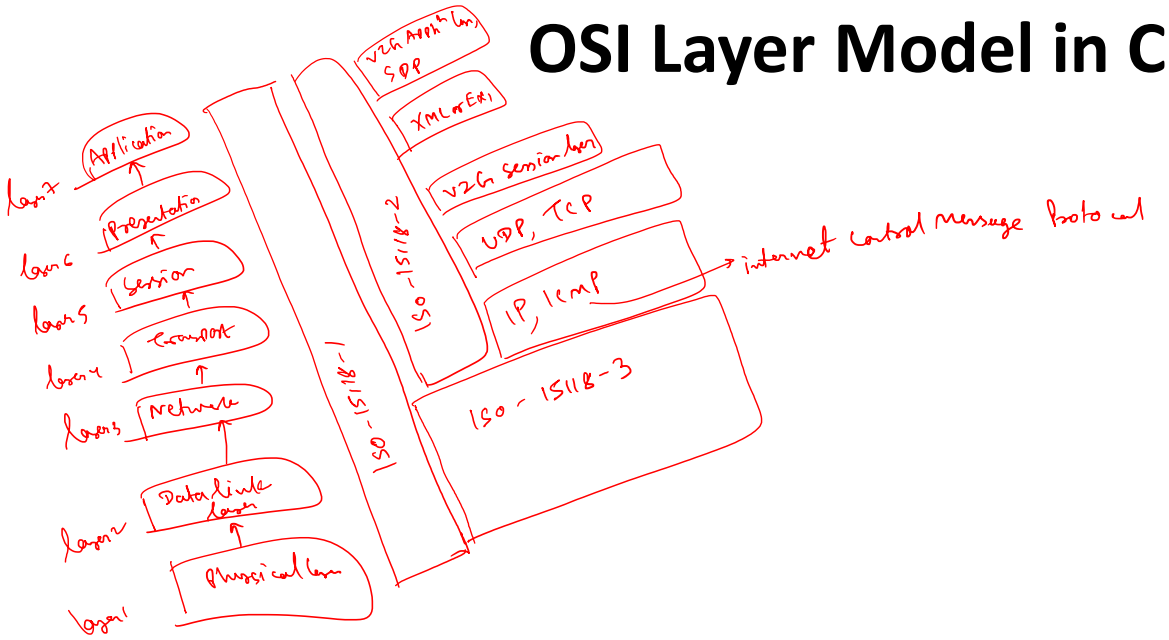
7 OSI Layer

(7) Physical layer → Signals
- air (WiFi) → radio signals
- Copper wire → electrical signals
- optical fibre → light signal





OSI Layer Model in CCS2



VDP – User datagram Protocol
TCP – Transmission Systems Protocol
XML – Extensible Markup Language
EXI – Efficient XML Interchange
SDP → SECC Discovery Protocol

Source:
https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

CCS2- HLC: OSI Layer1 (Physical)

- Included in ISO 15118 standard

Initial Condition	Function	Outcome
• PE & Control Pilot connected • PWM of 3-7%; State: "B2" • On both sides: compatible Home Plug Green PHY modems	Establish physical link on the PE-CP wires to the opposite side	PLC module ready for communication with a established frequency band of 2 MHz to 28 MHz

Layer ensures the start of physical connections to provide data transfer between EV and EVSE, also defines physical properties of the physical media such as voltage levels, frequency, etc.

Source:
https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

CCS2- HLC: OSI Layer2 (Datalink)

- Included in ISO 15118 standard

Initial Condition	Function	Outcome
Successfully established Physical layer	• Configuration of PLC nodes • Signal Level Attenuation Characterization (SLAC) procedure • Set-up logical connection of PLC Modems	Establishment of communication Link

Layer ensures an error-free transfer of data frames from one node to another over the physical media

Source:
https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

CCS2- HLC: OSI Layer2 (Network)

- Included in ISO 15118 standard

Initial Condition	Function	Outcome
Successfully established datalink layer	• Implement the IPv6 Internet Protocol Standard • Ensure unique addresses • Implementation of ICMPv6 to send error messages and SLAAC procedure	All nodes obtains the valid IP address

Layer controls the routing of connections which decides the path for the data which it should take

Source:
https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

Thank You